

Hierarchical Feature Recovery for Robust Human Activity Recognition in Body Sensor Networks

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ABSTRACT

With the advances in Body Sensor Networks (BSNs) and textile-integrated sensing, more sensor data becomes available from which human activities are recognised. However, some sensors may become unavailable unexpectedly in practice. Previous work proposed to complement the features of a missing sensor with regression-based methods but considered only up to one sensor missing and thus lacked a mechanism for selecting relevant sensors when multiple sensors were missing. The number of unique combinations of missing sensors increases exponentially when multiple sensors may be missing. To handle this, we propose a Hierarchical Feature Recovery (HFR) approach. We first assess the dependencies between sensors by comparing the feature mapping accuracy between each sensor and then evaluate the HFR approach on a dataset of activities of daily living with 17 gestures using 14 motion sensors. Our HFR method can alleviate classification performance drop by up to 8.3 pp compared to a baseline method.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing design and evaluation methods; • **Computer systems organization** → Sensor networks; Fault-tolerant network topologies.

KEYWORDS

Sensor-based Human Activity Recognition; Body Sensor Networks; Fault-tolerance; Robustness

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1 INTRODUCTION

Body Sensor Networks (BSNs) have many applications in health-care and assisted living [3]. Nowadays, motion sensors, e.g. Inertial Measurement Units (IMUs) and accelerometers, are embedded not only in smartphones [6] and smartwatches [17] but also in many other devices such as earbuds [5], eyeglasses [4], and rings [11]. Furthermore, smart textiles or e-textiles, i.e. fabrics that have integrated electric components, have been actively researched and developed in recent years [13]. These devices can be networked in a body area network; and the data from the network can be used to infer important elements of the user's contexts, such as human activities typically including gestures and daily routines, using machine learning techniques [1]. Activity recognition has important applications in sports, rehabilitation, personalised healthcare and a variety of industrial applications.

In this study, we are motivated by a common challenge encountered when using many networked wearable devices, which may not always be available. Wearers may not carry certain devices at all times, depending on their preferences, devices may run out of battery, and sensor failures may occur, e.g. textile-integrated sensors still face challenges of chemical and mechanical robustness when sensorized garments are washed or worn [9]. Even though most machine learning models for recognising human activities assume that all the sensors used for training the model are available at the time of inference, the same set of sensors are not always available in practice due to above-mentioned reasons. Therefore, BSN systems should be capable of handling these dynamically changing situations.

We present a method to flexibly deal with such situations wherein available sensors may dynamically change after deployment to improve Human Activity Recognition (HAR) robustness in BSN. Our approach aims to emulate the features which would be calculated on the missing sensor data by a process of regression from the features calculated from the data of available sensors. The regression models are learned at design time. At run-time, the appropriate regression models are selected based on the current state of the BSN. One key novelty of this work is to significantly reduce the number of regressors through a hierarchical feature recovery approach.

The contributions of this work are:

- A Hierarchical Feature Recovery (HFR) method that can efficiently recover features and address the exponential complexity explosion of previous approaches when multiple sensors are missing (Section 3).
- Evaluation of feature mapping accuracy between the features from the sensors used in the OPPORTUNITY dataset [2] to find feature dependencies, displaying which sensors are more important to complement the features of certain missing sensors (Section 4.1).
- Evaluation of our HFR method on the OPPORTUNITY 17 gesture classification task, showing that HFR can alleviate classification performance drop by 5.3 pp on average and up to 8.3 pp compared to a baseline method (Section 4.2).
- Discussion on possible improvements for feature calculation and hierarchical feature dependency graph for HFR and future work (Section 5, 6).

2 RELATED WORK

Typical Human Activity Recognition Chain (ARC) consists of sensor data acquisition, signal preprocessing, segmentation, feature extraction, and classification [1]. Missing sensor complementation in sensor-based human activity recognition is generally performed in either or combinations of the following three different phases in ARC: 1. raw sensor level, 2. feature level, and 3. classifier level.

Raw sensor level complementation focuses on temporal relationships of raw sensor data and estimates the values of missing sensors by using the raw sensor values before and/or after the missing values. Uchida et al. proposed to use the Autoregressive-Autoregressive (ARAR) algorithm to complement the raw sensor readings of missing sensors and showed that this method can suppress the decrease in human activity classification accuracy for short periods of missing sensor data up to 5.16 seconds [14].

Feature level complementation estimates the missing sensor features from the features of the other available sensors by using the learned relationships between features. Uchida and Ohmura demonstrated that multiple linear regression and kernel regression can be used to complement missing features and maintain higher accuracy than using classifiers trained only with the other available sensors in a situation where one accelerometer sensor out of six is missing [15].

Classifier level complementation estimates classifier outputs corresponding to missing sensors. Sagha et al. used the correlation among outputs from multiple classifiers to infer missing classifier outputs based on conditional Gaussian distribution [10]. Transition probabilities between activities can also be used to estimate missing classifier outputs [7].

These methods have pros and cons and can be used in combination. Uchida et al. compared their regression model-based feature level complementation method with the ARAR algorithm-based raw sensor level complementation method and showed that their feature level complementation method performed better where data loss is longer than two windows length long whereas the ARAR algorithm-based method performed better with shorter data loss [14]. Ohmura et al. have reported that better fault tolerance in human activity recognition can be achieved by combining data complementation methods at these different levels [7].

In the previous studies discussed above [7, 14, 15], feature-level complementation has only been tested in simplified situations where only one sensor is missing, assuming all the other sensors are available, and regression models for estimating the missing features are trained using all the available sensors. This approach might be feasible if we can assure that there can be only one missing sensor at a time as the number of possible patterns of missing sensors to consider is equal to the number of sensors used $\binom{n}{1} = n$, thus requires to train regression models for the n different patterns. However, when considering situations where up to $n - 1$ sensors may be missing, the number of combinations of missing sensors increases exponentially to the number of sensors used as in $\sum_{k=1}^{n-1} \binom{n}{k} = 2^n - 2$. Existing methods lack a mechanism for selecting relevant sensors for feature-level complementation in such situations.

3 HIERARCHICAL FEATURE RECOVERY

In this paper, we propose a Hierarchical Feature Recovery (HFR) method for complementing the features of missing wearable motion sensors. This method uses a hierarchical feature dependency graph, which makes BSN systems capable of efficiently handling situations where multiple and different sensors are missing by reducing the number of regression models required to handle all the possible combinations of missing sensors.

3.1 Hierarchical Feature Dependency Graph

A hierarchical feature dependency graph is a directed acyclic graph that describes hierarchical dependencies between the features calculated from wearable motion sensors. Relevant features are grouped hierarchically by inserting virtual feature nodes. It is important to note that the design of a hierarchical feature dependency graph should vary depending on the sensors used and their placement and the requirements on recognition accuracy and computational resource. In this study, we only consider the dependencies of the features from wearable motion sensors placed on the upper body. Our design of hierarchical feature dependency graphs in this study is inspired by the kinetic chain. Steindler defined the kinetic chain as a “combination of several successively arranged joints constituting a complex motor unit” [12]. Most upper extremity movements in Activities of Daily Living (ADLs) are open-chain movements in which their hands move freely.

Fig. 1 illustrates three different movements of a simplified model of the right arm with motion sensors attached (a) and its corresponding hierarchical feature dependency graph (b). In an Open Kinetic Chain (OKC) system, the movement of a segment propagates from proximal to distal, but not the other way around. For example, when the upper arm moves, the successive segments also move as in (3) in Fig. 1a, but the movement of the lower arm does not affect the movement of the upper arm as in (2) in Fig. 1a. In this regard, motion sensors attached to the upper arm are not capable of capturing movements that are specific to the lower arm (below the elbow joint) while upper arm movement triggers whole arm movements. Likewise, motion sensors attached to the forearm cannot tell the detailed movement of the right hand, but forearm movements accompany right hand movements. Therefore, in the feature dependency graph in Fig. 1b, features from sensors attached to the same bony segment are firstly grouped together; then, the

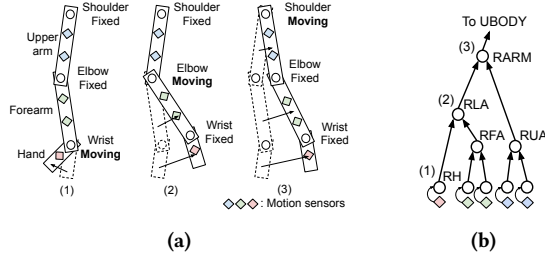


Figure 1: (a) Different movements of the right arm in an OKC system and (b) its corresponding feature hierarchical dependency graph (RH: Right Hand, RFA: Right Forearm, RLA: Right Lower Arm, RUA: Right Upper Arm, RARM: Right ARM, and UBODY: Upper BODY). The leaf nodes in (b) are the features calculated from the corresponding raw sensor data, and the non-leaf nodes are the higher-level features calculated from their dependent features. The numbers in brackets in (a) and (b) correspond to the movement and the features that capture the movement.

features from adjacent segments are stepwisely grouped together from distal to proximal; and lastly, the features from the right arm are grouped together with the features from the torso and the left arm to form the features of the upper body.

3.2 Feature Recovery Algorithm

This feature recovery algorithm aims to resolve the complete features at the root node of a hierarchical feature dependency graph, which is to be fed into a classifier.

Algorithm 1 ResolveFeatures

```

1: function RESOLVEFEATURES(node)
2:   if node.is_leaf then
3:     return node.features           ▶ return null when the
                                     corresponding sensor is missing
4:   end if
5:   for c_node in node.children do
6:     if c_node.features == null then
7:       c_node.features = ResolveFeatures(c_node)
8:     end if
9:   end for
10:  if all(node.children.features != null) then
11:    f = get_feat_builder(node)
12:    node.features = f.transform(node.children)
13:  else if any(node.children.features != null) then
14:    g = get_reg_model(node, node.avail_children)
15:    node.features = g.predict(node.avail_children)
16:  else
17:    node.features = null
18:  end if
19:  return node.features
20: end function

```

Algorithm 1 is the pseudo-code of our feature recovery algorithm. Suppose we have a predefined feature dependency graph, e.g. Fig.

2b. Each node object has its features and references to its child nodes. Leaf nodes have their features calculated from the raw sensor data unless those sensors are not available. For non-leaf nodes, the features are set to null by default. The function *ResolveFeatures* checks whether it has available features for each child node. If the features of the child node are not available, *ResolveFeatures* calls itself recursively passing the child node as the argument (lines 5 to 9). The main part of *ResolveFeatures* is from the line 10 to 19. It executes different operation according to the following three conditions:

- Condition 1.** if all child nodes' features are available, compute its features from the child node's features using a pre-defined feature calculation method.
- Condition 2.** if some child nodes' features are available, estimate its features from the available features using a pre-trained regression model;
- Condition 3.** if all child nodes' features are missing, make itself unavailable (set its features as null).

For Condition 1, a certain feature calculation method from child features needs to be defined in advance for each non-leaf node. The method could be a clustering method such as feature agglomeration or feature selection methods to reduce some redundancies in the child nodes' features, or just combining the child features in the simplest case. It takes its child nodes' features as input and calculates the target node's features. For Condition 2, regression models that estimate the target features from the available child features need to be prepared for each non-leaf node. If we denote the indegree of a node $v \in V$ by $deg^-(v)$, the number of regression models that needs to be trained for the node is $2^{deg^-(v)} - 2$. Therefore, $\sum_{v \in V} (2^{deg^-(v)} - 2)$ regression models are to be trained in total at design time. In both cases, the output shape of the features must be the same for each node.

4 EVALUATION

We have conducted two experiments: the first one is to investigate the relationships between wearable motion sensors placed at different body parts by comparing the feature mapping accuracy for each pair of the sensors; the second is to verify the effectiveness of HFR approach on a gesture classification task where multiple sensors are missing.

OPPORTUNITY Activity Recognition Dataset [2] is selected for both experiments. The dataset contains indoor human activities recorded in a sensor-rich environment which contains in total of 72 sensors of 10 different modalities, in the environment, on objects, and on the body. The data is collected from four subjects. Each subject performed five ADL sessions and one Drill session which contains repetitions of predefined activities. The dataset is annotated at three different levels: modes of locomotion (e.g. sitting, standing, walking), actions of the left and right hand (e.g. open door), and high-level activities (e.g. prepare sandwich).

In our experiments, we used the same dataset split used in the OPPORTUNITY challenge [2] for training and testing both regression and classification models: training with all ADL sessions of Subject 1, and ADL 1-3 and Drill sessions of Subject 2 and 3; testing with ADL4 and ADL5 of Subject 2 and Subject 3. We only considered IMUs and accelerometer sensors placed on the upper-body used

in the experiments, and excluded gyroscope and magnetometer channels of IMU sensors to align the modality between the sensors used in the experiments. The sample rate of these sensors is 30 Hz. Fig. 2a shows the placement of the sensors used in our experiments. For preprocessing, we applied linear interpolation to fill temporal missing values that happened during data collection. Then we applied 500 ms sliding window with a step size of 250 ms, followed by feature calculations of mean, median, standard deviation and range (max-min) for each channel, resulting in 12 features for each sensor.

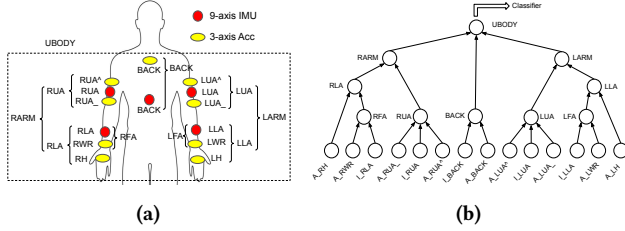


Figure 2: (a) Sensor placement of the 5 IMU sensors and 9 accelerometer sensors on the upper body in OPPORTUNITY Activity Recognition Dataset and (b) the corresponding Hierarchical feature dependency graph used in the experiment in Section 4.2. The prefixes 'A_' and 'I_' of the leaf nodes indicate the sensor type Accelerometer or IMU respectively. The first letter R or L means Right or Left.)

4.1 Sensor-to-Sensor Feature Mapping

In this experiment, we tested how accurately feature values could be complemented from one sensor to another sensor using the 14 sensors illustrated in Fig. 2a. The random forest regressor was used as the regression model. The max depth of the tree was set to 40 and the minimum number of samples required to be at a leaf node was set to 10. For the rest of the parameters, the default parameters of *scikit-learn* [8] implementation were used. Mean Absolute Error (MAE) was used to assess the mapping accuracy. To address the differences in the scales of the features, the values of each feature were normalised between 0 and 1 by min-max normalization before calculating MAE.

Fig. 3 summarizes the results of the feature mapping accuracies. It can be seen that the feature values of the sensors on the same segment or the same limb were estimated relatively well compared to the other combinations. In other words, sensors on different limbs or segments tend to be less relevant than those on the same limb or segment.

Comparing the features of the right arm with those of the left arm, the features of the sensors on the left arm were recovered with relatively better accuracy than those on the right arm. This is probably due to the fact that most of the gestures were performed with the right hand, and the left hand was relatively inactive compared to the right hand, thus making the left hand features more correlated with each other.

We can also see the tendency that mapping between the same type of sensors (IMU to IMU or Acc to Acc) and sensors on the mirrored positions on the left and right sides tend to perform better.

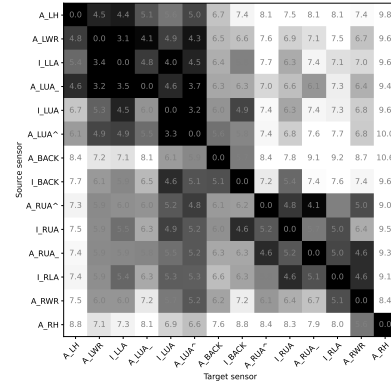


Figure 3: Summary of the feature mapping accuracy between sensors used in the dataset. Vertical axis is the source sensor, and horizontal axis is the target sensor. The values are MAE (multiplied by 10^2), i.e. the lower the value (the darker the colour), the more accurate the mapping.

The low accuracies of the mapping from A_RH and to A_RH is likely due to a large number of missing values during data collection.

4.2 Gesture Classification

In this experiment, we validated our HFR method by evaluating the performance on the OPPORTUNITY gesture classification task in situations where 0 to 13 sensors out of 14 are missing. We used the hierarchical feature dependency graph illustrated in Fig. 2b. For this experiment, we simply combined features from child nodes for Condition 1 in which all the child nodes' features were available, e.g. if a node has three child nodes and each of the three child nodes has 12 features, the parent node has 36 features. For Condition 2 in which some of the child node features were unavailable, we initially trained random forest regressors for each combination and used the regressors for feature complementation. We trained random forest regressors for all the $32 = 6 * (2^2 - 2) + 3 * (2^3 - 2)$ possible combinations of missing leaf nodes in the hierarchical feature dependency graph we used, which is significantly less compared to existing approaches $2^{14} - 2$. The parameters for the random forest regressors were the same as in the mapping analysis experiment. The random forest classifier was used as the classifier of the gestures. The max depth of the tree was set to 20 and the minimum number of samples required to be at a leaf node was set to 10. Only one classifier was trained using all the training data, and the complemented features using our HFR method were used as input for the evaluation. Classification performance was evaluated using the weighted f1-score to take class imbalances into account. For the baseline method, we simply filled missing values with zeros and used them as input to the classifier.

Fig. 4 shows the transitions of the weighted f1-score with k missing sensors ($0 \leq k < 14$). The line with circle markers shows the results with our HFR method, and the line with cross markers shows the baseline results. Compared to the baseline results, the recognition accuracy is almost consistently better, except for some cases in $k = 1$ and 13. With HFR, even with four sensors missing, the average accuracy is kept above 98% of the best accuracy when

no sensor is missing ($k = 0$). The differences in classification performance between HFR and the baseline are 5.3 pp on average and 8.3 pp at most.

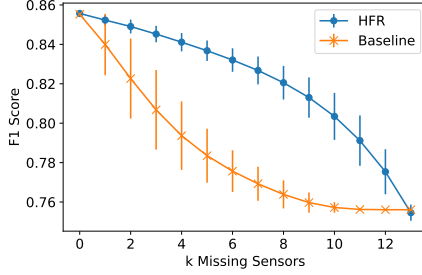


Figure 4: Comparison of the F1 scores obtained when k sensors are missing with HFR and the baseline method. The error bars are standard deviations. The performance obtained with our method (HFR) shows a supralinear characteristic, which indicates the benefit of this approach.

Fig. 5 shows a scatter plot with the results of our HFR method on the vertical axis and the baseline results on the horizontal axis for each missing sensor combination. This figure also shows that HFR method outperforms the baseline in most cases. In the mapping experiment, it was shown that the features of the right hand sensors are more difficult to complement than those of other locations (Fig. 3). This can explain why the classification performances were comparably lower in the dots in the circle. The results suggest that not just the number of missing sensors but also the pattern in which they are missing affects the classification accuracy.

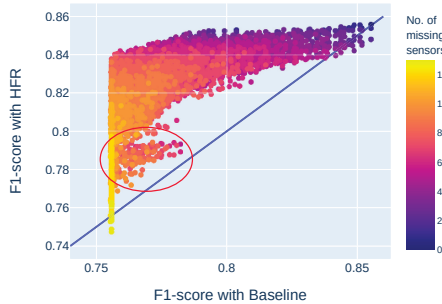


Figure 5: F1-score comparison between our HFR method and the baseline method for each combination of missing sensors. The color corresponds to the number of missing sensors k . Each dot represents the result with a particular combination of missing sensors. The further up a dot is from the linear line, the greater the benefit of our HFR method in that particular set of missing sensors. Most of the cases in the red circle are of all the right hand sensors missing.

The variance of the classification performance became larger as the number of missing sensors increased (Fig. 4 and 5), but for some combinations they worked fairly well even with more than half of the sensors being missing. To identify in which cases our method

brought the maximum benefits, we picked the top-5 combinations that were most distant from the linear line in Fig. 5 for each of the number of missing sensors $k = 9$ to 12, i.e. the number of available sensors is 2 to 5, and summarized in Table 1. When only two sensors were available, one sensor from each of the right and the left arm was available in all the top-5 cases. When the number of the available sensors was 3, 4 or 5, sensors were available from the right arm, left arm and back in most cases in a balanced manner. For the only combination without a back sensor, I_LUA and I_RUA were in the available sensors which had good mapping accuracy in the mapping analysis experiment (Fig. 3), suggesting that the BACK features could be complemented by these sensors to some extent.

Table 1: Top-5 combinations of available sensors that have most benefited from HFR method where k sensors are missing ($k = 9$ to 12), i.e. only $14 - k$ sensors are available. The combinations with the largest gaps in f1-score between HFR and the baseline were chosen. Each combination of available sensors is surrounded by parenthesis.

k	Top-5 combinations of available sensors
12	(A_LUA_, A_RUA_), (A_RWR, A_LUA_), (A_LUA_, I_RUA), (A_LUA_, I_RLA), (A_LWR, A_RUA_)
11	(I_BACK, A_LUA_, I_RLA), (I_BACK, A_LUA_, A_RUA_), (A_LUA_, A_BACK, I_RLA), (I_BACK, A_LWR, I_RLA), (A_LUA_, I_RUA, I_LUA)
10	(I_RUA, I_LUA, A_LUA_, I_RLA), (A_RUA_, I_RUA, I_LUA, A_LUA_), (A_RWR, A_LWR, I_BACK, I_RLA), (A_RWR, I_BACK, A_LUA_, I_RLA), (I_LUA, I_BACK, A_LUA_, I_RLA)
9	(I_LUA, A_RWR, I_BACK, A_LUA_, I_RLA), (A_RUA_, I_RUA, I_LUA, I_BACK, A_LUA_), (I_RUA, I_LUA, I_BACK, A_LUA_, I_RLA), (A_RUA_, I_RUA, I_LUA, A_LUA_, I_RLA), (I_RUA, I_LUA, A_RWR, I_BACK, A_LUA_)

5 DISCUSSION

The results of the gesture classification experiment in Section 4.2 demonstrate that our HFR method can effectively mitigate the decrease in the recognition performance when multiple sensors are missing while significantly reducing the number of combinations that regression models needed to be trained for compared to existing approaches from $2^n - 2$ to $\sum_{v \in V} (2^{\deg^-(v)} - 2)$; n is the number of sensors and V is the set of nodes in the feature dependency graph. In the case of our experiment with OPPORTUNITY dataset using 14 wearable motion sensors attached to the upper body, the number of combinations reduced from $2^{14} - 2$ to 32 ($= 6 * (2^2 - 2) + 3 * (2^3 - 2)$).

In this study, we specifically focused on feature-level complementation, assuming that particular sensors may be unavailable unexpectedly for a relatively long time, during the whole evaluation time in our case. However, as mentioned in Section 2, complementation methods at different levels can be used in combination. For example, if some sensors are missing for only a short period of time,

e.g. up to a few seconds, raw-sensor level complementation methods such as ARAR algorithm [14] could be used. Also, in situations where classifier-level complementation is taken place, our method could be used in combination to complement missing features associated with each classifier if a part of the features is available.

However, there is still room for additional investigation on how to calculate features at each node and how to design feature dependency graphs. Regarding feature calculation at each node, there are two aspects that should be considered: 1. features should be informative, i.e. they contribute to better classification accuracy, and 2. features should be recoverable when some of the child nodes are missing. We only calculated four simple features from raw sensor data, namely mean, median, standard deviation, and range; therefore, other sets of features should be explored to examine if our HFR method still works with them. Wang et al. have proposed a large set of features (2,727 in total) in the time-domain and frequency-domain for human activity recognition with a feature selection method based on mutual information [16]. In addition, we defined non-leaf nodes' feature sets by simply concatenating the features of their child nodes. These features from child nodes may contain redundant features especially those from sensors attached to the same bony segment, thus feature selection will also be needed where more features are used. Besides, it is desirable that non-leaf node features are recoverable from any set of available child nodes and not heavily dependent on their specific child node(s). Therefore, the recoverability of the features should also be considered when applying feature selection. Furthermore, careful consideration should be given to whether it is necessary to forcefully recover features where the mapping is known to be inaccurate. For hierarchical feature dependency graphs, the one we used in the experiment was designed with reference to the concept of kinetic chain, which successfully reduced the number of combinations of regression models needed while mitigating the decrease in the recognition performance. However, we acknowledge that this is not the only way of designing a hierarchical feature dependency graph. Moreover, this graph is not optimized either in terms of computational efficiency or in terms of activity recognition accuracy. Therefore, optimization methods for improving hierarchical feature dependency graphs based on different objectives are desirable as well as ways to automatically construct dependency graphs from sensor-to-sensor feature mapping results such as Fig. 3. For example, child nodes can also be added to increase the accuracy of regression of a certain node. Fig. 3 shows that the features of I_BACK sensor can be well mapped from the features of I_LUA and I_RUA, hence arcs to I_BACK from I_LUA and I_RUA could be added to the feature dependency graph in Fig. 2b. Also, the number of regression models that need to be trained for a node can be reduced by reducing the number of its children, i.e. it becomes minimum when the feature dependency graph is a binary tree.

Our method only requires a hierarchical feature dependency graph defining the relationships between the sensors and those sensors' data collected at the same time to learn the mappings between features. This process is done at training time and is not dependent on specific features or machine learning algorithms. Therefore, our approach can be added without affecting the rest of the pipeline in ARC.

6 CONCLUSIONS

We proposed HFR: a method for complementing missing features with regression using a hierarchical feature dependency graph to improve HAR robustness in BSN against missing sensors. HFR can deal with situations where multiple wearable motion sensors are missing and addresses the exponential complexity explosion of the previous approaches [7, 14, 15]. We firstly investigated the relevance between sensors attached to different body parts by comparing the accuracy of feature regressions from one sensor to another using 14 on-body motion sensors, 5 IMUs and 9 accelerometers, attached to the upper body. We then validated the performance of the proposed method on the OPPORTUNITY gesture classification task while changing the number of missing sensors from 1 to 13. The hierarchical feature dependency graph we used in this study was designed based on the concept of kinetic chain. The results showed that our HFR method mitigates the decrease in gesture classification performance compared to the baseline method which fills missing values with zeros. In the future, we will investigate how to find better feature sets and design feature dependency graphs depending on the task, specific sensors used, and their placement.

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